INSA de Toulouse
Département GMM
BE - Processus de Poisson et Application en actuariat et fiabilité - 4 MA
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The Cramér-Lundberg modelling of risk in non-life insurance

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1 Context of the project

The main goal of this project is to study a modeling of wealth/risk in actuarial sciences called the Cramér-Lundberg model. More precisely, the wealth of an insurance company can be modeled in a simple way as the difference of inwards (represented by the parameters $u, p \geq 0$) and outwards cashflows involving a compound Poisson process of the following form

$$R_t = u + pt - \sum_{i=1}^{N_t} Y_i, \quad t \geq 0; \tag{1}$$

where N is a Poisson process with constant intensity $\mu > 0$ and where the non-negative iid random variables $(Y_i)_{i\geq 1}$ represent the values of the sinisters involved in the contract; $u \geq 0$ is an initial capital; and $p > 0$ represents the premium rate.

The goal of this project is twofold:

- Investigate the properties of this model in terms of risk analysis, that is given μ and a choice of claims $(Y_i)_i$ what should be the values of the the premium u and p ?
- Compare the classical Cramér-Lundberg model (1) with a slightly different model that will be made precise below.

2 Work to be performed

For each of the two sections below, each group delivers a Notebook Python file addressing the theoretical and numerical elements listed below. Questions regarding the code will be asked during the defense.

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2.1 Risk analysis in the Cramér-Lundberg model (1)

In this section we assume that $(Y_i)_i$ are iid random variables valued in $\mathbb{R}_+$ with $m := \mathbb{E}[Y_1] < +\infty$, N is an homogeneous Poisson process with intensity $\mu > 0$.

1. Give an explicit form for $\mathbb{E}[R_t]$ for any time t.

2. Give the value p^* such that for any $p > p^*$, $\mathbb{E}[R_t] > 0$ for any t. Does this value depend on u ?

3. In practice the regulator is interested in computing the so-called ruin probability over the time of the contract that is:

$$\mathbb{P}[\exists t \leq T,\ R_t < 0].$$

   To simplify our analysis we study a simplified approximated notion, and consider the Bernouilli random variable $\mathcal{R}$ as:

$$\{\mathcal{R} = 1\} \Leftrightarrow \{R_T < 0\}.$$

   Fix $T = 100$ and choose $\mu = 1$, $Y_i := \mathcal{E}(\beta),\ \beta = 0.1$, $p = (1+\theta)p^*$ with $\theta = 30\%$. All these parameters are fixed now and only the impact of the u parameter is studied.

   (a) Compute m.

   (b) Using the Monte-Carlo algorithm, compute for each value of u in $\{10k,\ k \in \{1, \ldots, 20\}\}$,

$$u \mapsto \mathbb{P}[\mathcal{R} = 1].$$

   (c) Fit the curve to match with $\exp(-\kappa u)$ for some κ that you determine.

   (d) Give an interpretation in terms of risk-investment analysis of this behavior.

2.2 Opening

Consider an alternative model in which claims are discounted with respect to a rate ρ. More precisely consider an alternative model of the form:

$$R_t = u + pt - \sum_{i=1}^{N_t} e^{-\rho(t-\tau_i)} Y_i, \quad t \geq 0; \tag{2}$$

where $\tau_i := \inf\{t > ,\ N_t = i\}$ and $\rho \geq 0$ is a discount factor.